2.17

(a) $D = -\frac{1}{4\mu_0} D(F^{\alpha\beta}F_{\alpha\beta}) - KA_\alpha J^\alpha$

$$S = \frac{1}{c}\int d\Omega \; D \Rightarrow \delta S = -\int \frac{d\Omega}{c}\,\delta\left[\frac{1}{4\mu_0} D(F^{\alpha\beta}F_{\alpha\beta}) + KA_\alpha J^\alpha\right]$$

where the variation is with respect to A_α, J^α is kept constant.

$$\delta D(F^{\alpha\beta}F_{\alpha\beta}) = D'(x)\left(\frac{\partial x}{\partial F^{\alpha\beta}}\delta F^{\alpha\beta} + \frac{\partial x}{\partial F_{\alpha\beta}}\delta F_{\alpha\beta}\right) =$$

$=$ (where $x = F^{\alpha\beta}F_{\alpha\beta}$ is the argument of Lagrangian D.)

$$= D'(x)\left(F_{\alpha\beta}\delta F^{\alpha\beta} + F^{\alpha\beta}\delta F_{\alpha\beta}\right) = 2D'(x)\, F_{\alpha\beta}\delta F^{\alpha\beta}$$

$$= 2D'(F^{\alpha\beta}F_{\alpha\beta})\underbrace{F^{\alpha\beta}\delta F_{\alpha\beta}} = -4D'(F^{\alpha\beta}F_{\alpha\beta})\, F^{\alpha\beta}\frac{\partial \delta A_\alpha}{\partial x^\beta}.$$

$$\underset{\text{antisymmetric}}{F^{\alpha\beta}}\underbrace{\left(\frac{\partial \delta A_\beta}{\partial x^\alpha} + \frac{\partial \delta A_\alpha}{\partial x^\beta}\right.}_{\text{symmetric tensor}}\left. - 2\frac{\partial \delta A_\alpha}{\partial x^\beta}\right)$$

Thus $\delta S = \int \frac{d\Omega}{c}\left[-\frac{(-4)}{4\mu_0} D'(x)\, F^{\alpha\beta}\frac{\partial \delta A_\alpha}{\partial x^\beta} - KJ^\alpha\delta A_\alpha\right]$

need to integrate by parts

full derivative $\Rightarrow$ (hyper)surface contribution $\Rightarrow$ gives $\emptyset$

$$= \int \frac{d\Omega}{c}\left\{\frac{\partial}{\partial x^\beta}\left(\frac{1}{\mu_0}D'(x)\, F^{\alpha\beta}\,\delta A_\alpha\right) - \frac{\partial}{\partial x^\beta}\left(\frac{1}{\mu_0}D'(x)F^{\alpha\beta}\right)\delta A_\alpha - KJ^\alpha\delta A_\alpha\right\}$$

$$= \int \frac{d\Omega}{c}\left\{-\frac{\partial}{\partial x^\beta}\left(\frac{1}{\mu_0}D'(x)\, F^{\alpha\beta}\right) - KJ^\alpha\right\}\delta A_\alpha = 0$$

$$\Rightarrow \frac{\partial}{\partial x^\beta}\left(D'(F^{\alpha\beta}F_{\alpha\beta})\, F^{\alpha\beta} \right) = -\mu_0 K J^\alpha$$

If $D'(x) \to K$ for $x \to 0$, then we recover Maxwell's eqn

$$\frac{\partial F^{\alpha\beta}}{\partial x^\beta} = -\mu_0 J^\alpha$$

(Note that eq.(2.183) should have minus sign (−).)

(b) For a point charge at the origin $\rho(\vec{r}) = q\,\delta(\vec{r})$

$$J^\alpha = (c\rho, 0) \quad , \quad F^{0\beta} = (0, -E_x/c, -E_y/c, -E_z/c)$$

$$\cancel{\frac{\partial}{\partial x^\beta}(F^{0\beta}) = -\frac{1}{c}\left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\right)}$$

$$\Rightarrow -\frac{1}{c}\frac{\partial}{\partial x^\beta}\left(D' E_\beta\right) = -\mu_0 K c \rho \quad (\text{here } \beta = 1,2,3)$$

$$\vec{\nabla}\cdot\left(D' \vec{E}\right) = \mu_0 c^2 K \rho = \frac{K\rho}{\varepsilon_0}$$

Integrate both sides over the volume:

$$\underbrace{\int dV\, \vec{\nabla}\cdot\left(D'\vec{E}\right)} = \frac{K}{\varepsilon_0}\int dV\, q\,\delta(\vec{r}) = \frac{Kq}{\varepsilon_0}$$

$$D'(F^{\alpha\beta}F_{\alpha\beta})\, E(r)\cdot 4\pi r^2 = \frac{Kq}{\varepsilon_0}$$

$$E(r) = \frac{q}{4\pi\varepsilon_0 r^2}\,\frac{K}{D'(F^{\alpha\beta}F_{\alpha\beta})}$$

(c) Born – Infeld proposed $D(x) = 4K \frac{E_0^2}{c^2}\left(\sqrt{1+\frac{c^2}{2E_0^2}X}-1\right)$

$$X = F^{\alpha\beta}F_{\alpha\beta}$$

Since $F^{\alpha\beta}F_{\alpha\beta} = 2\left(B^2 - \frac{E^2}{c^2}\right) \Rightarrow -\frac{2}{c^2}E^2$ since $\vec{B}=0$ for static point charge.

$$D'(x) = 4K\frac{E_0^2}{c^2}\,\frac{1}{2}\,\frac{1}{\sqrt{1+\frac{c^2X}{2E_0^2}}}\,\frac{c^2}{2E_0^2} = \frac{K}{\sqrt{1+\frac{c^2}{2E_0^2}X}}$$

$$D'(F^{\alpha\beta}F_{\alpha\beta}) = \frac{K}{\sqrt{1+\frac{c^2}{2E_0^2}\left(-\frac{2E^2}{c^2}\right)}} = \frac{K}{\sqrt{1-\frac{E^2}{E_0^2}}}$$

$$\Rightarrow E(r) = \frac{q}{4\pi\varepsilon_0 r^2}\,\frac{K}{K}\sqrt{1-\frac{E(r)^2}{E_0^2}}$$

$$E^2 = \left(\frac{q}{4\pi\varepsilon_0 r^2}\right)^2\left(1-\frac{E^2}{E_0^2}\right)$$

$$E^2\left[1+\left(\frac{q}{4\pi\varepsilon_0 r^2 E_0}\right)^2\right] = \left(\frac{q}{4\pi\varepsilon_0 r^2}\right)^2$$

$$E = \frac{q}{4\pi\varepsilon_0 r^2}\,\frac{1}{\sqrt{1+\left(\frac{q}{4\pi\varepsilon_0 r^2 E_0}\right)^2}}$$

$$\Rightarrow E(r\to 0) \to \frac{q}{4\pi\varepsilon_0 r^2}\sqrt{\left(\frac{4\pi\varepsilon_0 r^2 E_0}{q}\right)^2} = E_0 \quad !$$

2.18 $D = K_1 \varphi + K_2 \partial^\alpha \partial_\alpha \varphi + K_3 \varphi^2 + K_4 \partial^\alpha \varphi \partial_\alpha \varphi + K_5 \rho_\varphi \varphi$

K_{1-5} are constants, ρ_φ = scalar source density, keep constant during variation.

(a) $S = \frac{1}{c}\int d\Omega\, D$, $\delta S = \frac{1}{c}\int d\Omega\, \delta D$

$$\delta D = K_1 \delta\varphi + K_2 \delta(\partial^\alpha \partial_\alpha \varphi) + 2K_3 \varphi \delta\varphi + K_4 \delta(\partial^\alpha \varphi \partial_\alpha \varphi) + K_5 \rho_\varphi \delta\varphi.$$

$$\delta(\partial^\alpha \partial_\alpha \varphi) = \delta \frac{\partial^2 \varphi}{\partial x_\alpha \partial x^\alpha} = \frac{\partial^2 \delta\varphi}{\partial x_\alpha \partial x^\alpha}$$

The corresponding part of δS is $\delta S = \frac{1}{c}\int d\Omega\, K_2 \frac{\partial^2 \delta\varphi}{\partial x_\alpha \partial x^\alpha}$ and reduces to the (hyper)surface contribution on which $\delta\varphi = 0 \Rightarrow \delta(\partial^\alpha \partial_\alpha \varphi)$ does not contribute to δS at all.

Next $\delta\left(\frac{\partial \varphi}{\partial x_\alpha}\frac{\partial \varphi}{\partial x^\alpha}\right) = \frac{\partial \delta\varphi}{\partial x_\alpha}\frac{\partial \varphi}{\partial x^\alpha} + \frac{\partial \varphi}{\partial x_\alpha}\frac{\partial \delta\varphi}{\partial x^\alpha} =$

$$= 2\frac{\partial \varphi}{\partial x^\alpha}\frac{\partial \delta\varphi}{\partial x_\alpha} = \underbrace{2\frac{\partial}{\partial x_\alpha}\left(\frac{\partial \varphi}{\partial x^\alpha}\delta\varphi\right)}_{\text{gives surface contribution in } \delta S \Rightarrow \underline{\text{zero}}} - 2\frac{\partial^2 \varphi}{\partial x_\alpha \partial x^\alpha}\delta\varphi$$

$$\Rightarrow \delta S = \frac{1}{c}\int d\Omega \left(K_1 + 2K_3\varphi - 2K_4 \frac{\partial^2 \varphi}{\partial x_\alpha \partial x^\alpha} + K_5 \rho_\varphi\right)\delta\varphi = 0$$

$$\Rightarrow \left(-\frac{\partial^2 \varphi}{\partial x_\alpha \partial x^\alpha} + \frac{K_3}{K_4}\varphi\right)\cdot 2K_4 = -K_1 - K_5 \rho_\varphi$$

$$\Rightarrow \frac{\partial^2 \varphi}{\partial x_\alpha \partial x^\alpha} + \mu_\varphi^2 \varphi = \chi_0 \rho_\varphi$$

where $\mu_\varphi^2 = -\frac{K_3}{K_4}$, $\chi_0 \rho_\varphi = \frac{K_1 + K_5 \rho_\varphi}{2K_4}$

(Really K_1 should be ϕ, $K_1 = 0$, in order for this to make sense)

(b) So $\frac{\partial^2 \varphi}{\partial x_\alpha \partial x^\alpha} + \mu_\varphi^2 \varphi = \chi_0 \rho_\varphi$ ← scalar density

For a point charge in the rest frame $\rho_\varphi(\vec{r}) = q\,\delta(\vec{r})$.
The total charge $Q = \int dV\, \rho_\varphi(\vec{r}) = q$.
In any other inertial frames $Q = Q' = \int dV'\, \rho'_\varphi(\vec{r}') = q$
But $dV' = \gamma dV$

$dV' = dx'dy'dz'$, $dV = dxdydz$, $dx' = \gamma\, dx$

$\Rightarrow \rho_\varphi = \frac{\rho'_\varphi}{\gamma}$ for the moving charge

$$\frac{\partial^2 \varphi}{\partial x_\alpha \partial x^\alpha} + \mu_\varphi^2 \varphi = \chi_0 \frac{q}{\gamma} \delta(\vec{r} - \vec{r}_0(t))$$

2.19 <u>Scalar waves</u> $\frac{\partial^2 \varphi}{\partial x_\alpha \partial x^\alpha} + \mu_\varphi^2 \varphi = 0$ in a free space.

(a) $$\Rightarrow \frac{1}{c^2}\frac{\partial^2 \varphi}{\partial t^2} - \nabla^2 \varphi + \mu_\varphi^2 \varphi = 0$$

Plane wave $\varphi(r,t) \sim e^{i(\bar{k}\cdot\bar{x} - \omega t)}$

$$\left(-\frac{\omega^2}{c^2} + k^2 + \mu_\varphi^2\right)\varphi = 0$$

$\Rightarrow$ dispersion relation is $\omega = c\sqrt{\mu_\varphi^2 + k^2}$

Phase velocity $v_\phi = \frac{\omega}{k} = c\sqrt{1 + \frac{\mu_\varphi^2}{k^2}}$

Group velocity $v_g = \frac{d\omega}{dk} = \frac{ck}{\sqrt{\mu_\varphi^2 + k^2}} < c$. <u>okay</u>

(b) Stationary charge $\Rightarrow \gamma = 1$.

Time-independent solution $\varphi(\vec{r})$:

$$-\nabla^2 \varphi + \mu_\varphi^2 \varphi = \chi_0 q\, \delta(\vec{r}).$$

Let $\varphi(\vec{r}) = \int \frac{d^3\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot\vec{r}} \varphi(\vec{k})$, $\delta(\vec{r}) = \int \frac{d^3\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot\vec{r}}$

$$\Rightarrow -\nabla^2 \varphi = \int \frac{d^3\vec{k}}{(2\pi)^3} k^2 e^{i\vec{k}\cdot\vec{r}} \varphi(\vec{k})$$

$$\Rightarrow \int \frac{d^3\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot\vec{r}} (k^2 + \mu_\varphi^2)\, \varphi(\vec{k}) = \chi_0 q \int \frac{d^3\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot\vec{r}}$$

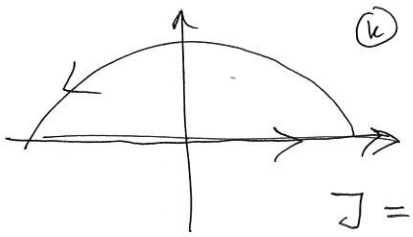
$$(k^2 + \mu_\varphi^2)\, \varphi(\vec{k}) = \chi_0 q \Rightarrow \varphi(\vec{k}) = \frac{\chi_0 q}{k^2 + \mu_\varphi^2}$$

$$\Rightarrow \varphi(\vec{r}) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{(\chi_0 q)\, e^{i\vec{k}\cdot\vec{r}}}{k^2+\mu^2} = \chi_0 q \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \cdot$$

$$\cdot \int_0^\infty \frac{dk\, k^2}{(2\pi)^3} \frac{e^{ikr\cos\theta}}{k^2+\mu^2} = \frac{\chi_0 q}{(2\pi)^2} \int_0^\infty dk \frac{k^2}{k^2+\mu^2} \underbrace{\int_{-1}^{1} d(\cos\theta)\, e^{ikr\cos\theta}}_{\left.\frac{e^{ikr\cos\theta}}{ikr}\right|_{-1}^{1}}$$

$$= \frac{\chi_0 q}{(2\pi)^2} \frac{2i}{ir} \int_0^\infty dk \frac{k \sin(kr)}{k^2+\mu^2} = \frac{\chi_0 q}{4\pi^2 r} \cdot J$$

Consider $J = \int_{-\infty}^{\infty} dk \frac{k \sin(kr)}{k^2+\mu^2} = \mathrm{Im} \int_{-\infty}^{\infty} dk \frac{k e^{ikr}}{k^2+\mu^2}$



$r>0 \Rightarrow$ can close contour in upper k-plane with $\mathrm{Im}(k)>0$: pole at $k = i\mu$.

$$J = \mathrm{Im} \int_{-\infty}^{\infty} dk \frac{k e^{ikr}}{(k-i\mu)(k+i\mu)} =$$

$$J = \lim \left\{ 2\pi i \; \frac{i\mu \, e^{ir(i\mu)}}{2i\mu} \right\} = \pi e^{-\mu r}$$

$$\Rightarrow \varphi(\vec{r}) = \frac{\chi_0 q}{(2\pi)^2} \frac{1}{r} \pi e^{-\mu r} = \frac{\chi_0 q}{4\pi} \frac{e^{-\mu r}}{r}.$$

This is Yukawa potential. It decays exponentially with distance on a scale $\frac{1}{\mu}$.